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Original Article

Reusable glycan microarrays using a microwave assisted wet-erase (MAWE) process

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Modern studies on binding of proteins to glycans commonly involve the use of synthetic glycans and their derivatives in which a small amount of the material is covalently printed onto a functionalized slide in a glycan microarray format. While incredibly useful to explore binding interactions with many types of samples, the common techniques involve drying the slides, which leads to irreversible association of the protein to the spots on slides to which they bound, thus limiting a microarray to a single use. We have developed a new technique which we term Microwave Assisted Wet-Erase (MAWE) glycan microarrays. In this approach we image the slides under wet conditions to acquire the data, after which the slides are cleaned of binding proteins by treatment with a denaturing SDS solution along with microwave treatment. Slides cleaned in this way can be reused multiple times, and an example here shows the reuse of a single array 15 times. We also demonstrate that this method can be used for a single-array per slide or multi-array per slide platforms. Importantly, the results obtained using this technique for a variety of lectins sequentially applied to a single array, are concordant to those obtained via the classical dry approaches on multiple slides. We also demonstrate that MAWE can be used for different types of samples, such as serum for antibody binding, and whole cells, such as yeast. This technique will greatly conserve precious glycans and prolong the use of existing and new glycan microarrays.

Key words: glycan binding proteins; glycan microarrays; microwave; reusable microarrays.

Introduction

Glycans are carbohydrate molecules found in all living organisms, often attached to proteins, lipids or other biomolecules (Varki 2017; Schjoldager et al. 2020). They elicit and regulate numerous biological functions such as cell adhesion, protein localization, immunity and infection (Taylor et al. 2022) by interacting with adhesion molecules, such as glycan binding proteins (GBPs) and with antibodies (Cummings et al. 2022). Therefore, studying the binding of GBPs or antibodies to libraries of glycans is of great importance. However, glycans are often structurally complex, making them difficult to synthesize and as a result are often expensive or challenging to obtain.

To facilitate the study of these precious glycans, researchers have developed microarray technologies wherein small (low nanogram to picogram) amounts of glycan derivatives, in many cases with primary amines for conjugation, are deposited onto a functionalized slide (usually glass) as individual spots and are typically covalently conjugated to the surface (Rillahan and Paulson 2011; Mehta et al. 2020). One of the most commonly used slide technologies is 3D-Hydrogel slides such as those manufactured by Schott (Nexterion Slide H) and others, which is a polymeric N-hydroxysuccinamide (NHS)-activated surface. The NHS surface reacts with the amines of the linker on the glycan derivatives to produce a covalent bond. These glycan microarrays can then be used to probe GBP or antibody interactions in a manner similar to enzyme-linked immunosorbent assays (ELISAs) (Blixt et al. 2004; Alvarez and Blixt 2006; Gao et al. 2019). For a GBP, for example, it may be incubated on the glycan microarray slide, followed by secondary detection reagents if the GBP is not directly fluorophore conjugated. The slides are then dried and scanned using a microarray scanner to obtain an image. The fluorescence intensity of the spots in the image are quantified, which is the measure of binding of the GBP to each individual glycan printed in distinct spots (Heimburg-Molinaro et al. 2011). While this classical method of performing glycan microarray studies has been extremely useful in studying thousands of GBP-glycan interactions, a key drawback of this approach is the lack of reusability of the glycan-printed microarrays. This means that a single microarray slide could typically be used only one time to study the binding of a single sample, and the binding on the array could not be removed without compromising the slide.

To overcome the hurdle of reusability of glycan microarrays, we explored several methods and conditions to devise a process by which we could successfully remove bound GBPs and re-utilize the same array. In this paper we provide a method termed Microwave Assisted Wet-Erase (MAWE) glycan microarray, which enables glycan arrays to be reused, without altering subsequent binding or introducing detectable artifacts. We discovered that the key components of reusability are maintaining slide hydration at all times, and using microwaves along with the denaturant SDS to remove

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binding. We showed that MAWE can be used to probe lectins (as mixtures or in isolation) on a single array robustly up to 15 times, in either single-array per slide (such as the CFG microarrays from the Consortium for Functional Glycomics) or multi-array per slide (such as the numerous types of NCFG microarrays at the National Center for Functional Glycomics) platforms. We further demonstrate the utilization of MAWE on diverse sample types such as human serum and yeast expressing lamprey variable lymphocyte receptors (VLRs). This method will help to dramatically increase the utility of glycan microarrays and conserve glycans which are difficult and costly to obtain.

Results

Wet scanning workflow and comparison with dry method

In our many years of experience, once a 3D-Hydrogel slide has been used in a microarray experiment to detect binding of either GBPs or antibodies, and the slide is subsequently dried with the sample overlayed, it appears so far impossible to remove the bound sample without compromising the integrity of the array. Thus, we sought to develop a workflow that allows the slides to be hydrated at all times (Fig. 1a), as opposed to drying the slides for the scanning step. The slides were kept hydrated at all times (even during data acquisition) using either a SecureSeal™ Hybridization Chamber or ProPlates® Multi-Well Chamber (1-well) (Grace Bio-Labs) for the CFG arrays (single-array per slide), or the ProPlates® Multi-Well Chamber (8-well) (Grace Bio-Labs) for the NCFG arrays. These chambers were ideal in balancing between preventing leaks of the buffers and avoiding air bubbles. We also tried thinner HybriWell™ Sealing System (Grace Bio-Labs), but the chambers were too small and resulted in air bubbles during handling.

For acquisition, the microarray scanner GenePix 4400A (Molecular Devices) that we typically use, requires the slide to be completely dry and also placed upside down. Furthermore, since microarray scanners use lasers which have limited focusing depth, they are unable to focus on the surface if there is any liquid or chamber attached. Since traditional microarray scanners require complete drying of the slides, we developed a workflow to use a digital microscope. We used the ImageXpress Pico (Molecular Devices) digital microscope and acquired a stitched image acquisition using the 2.5× (or sometimes 4×) objective. We chose this device as the files can easily be transferred from the software to the GenePix Pro software (same manufacturer for both instruments). The exposure time was optimized to 5 s to provide sufficient signal intensities. The image acquired by the digital microscope is 12-bit (max. intensity of 4,096) but can be exported as a 16-bit TIFF file (max. intensity of 65,536) for subsequent analysis using GenePix Pro software. This difference in bit count results in lower numerical intensity values, as the export only changes the file bit rate and not the values of the pixels, but does not affect overall binding patterns (see below). The array layout file, i.e. the .gal file, also needed to be scaled to the image, as the resolution (μm/pixel) is different between the digital microscope and the microarray scanners. With the scaled array layout file (GAL file), alignments were performed and quantified to produce the results file (GPR file), which can be analyzed using traditional glycan microarray workflows.

We wanted to ensure that results obtained using the wet method of scanning were comparable to the dry method. To test this, we applied a cocktail of vetted and well-documented biotinylated lectins (see below for full names): ConA, GNL, PHA-E, PHA-L, MAL-I, SNA, AAL, LCA, RCA-I, GSL-I, GSL-II, at 2 μg/mL, followed by streptavidin-Cy5 at 0.5 μg/mL onto a CFG microarray. Data was acquired under dry conditions using the GenePix scanner, or under the wet conditions using the ImageXpress Pico. As can be seen from Fig. 1b, the normalized binding intensity pattern between the two instruments remains highly correlated with an r-value (Pearson correlation) of 0.869 (Fig. 1c). In fact, closer examination reveals better relative binding for several glycans under wet conditions than dry (as indicated by the greater deviation from the diagonal of the data points towards the "wet" axis in the scatter plot). These results show favorable correlation between the classical dry and the new wet scanning methods.

To check the sensitivity of wet scanning in comparison to the dry scanning technique, we tested AAL lectin in a dilution series from 20 μg/mL to 0.02 μg/mL on the Lectin QA/QC array and scanned it first under wet conditions (using the ImageXpress Pico) and then under dry conditions (using the GenePix). We noticed that the wet scanning conditions produced a proportional but lower overall signal compared to the dry method of scanning (Supplementary Fig. S1a). However, when the data was scaled to the dataset max (Supplementary Fig. S1b), it can be seen that the overall binding patterns between the wet and dry conditions for all concentrations are similar. These results indicate that the perceived limitation is more due to the instrument capability rather than the technique, as the ImageXpress Pico uses an LED-based light source, in comparison to the more powerful laser-based light source found in the GenePix.

Optimization of wet erase parameters

We tried several conditions and methods to determine which method would be best at removing the binding of test materials, while still providing robust reusability (Supplementary Table S1). We discovered that repeatedly microwaving the slide in a regular microwave oven with sodium dodecyl sulfate (SDS) solution resulted in the best cleaning action, as well as allowing for robust reusability of the slide. Application of an SDS solution at room temperature (quick washes or long incubation) was not sufficient to remove binding, and neither was quick washes using hot SDS solution. Other denaturants such as guanidine hydrochloride or saturated NaCl solution did not sufficiently remove bound material. We avoided heating/microwaving guanidine hydrochloride, as it is unstable upon heating. To check if the microwave might alter the SDS solution, we tried pre-microwaving the SDS and then applying it, however, that did not reduce the binding. We then tried microwaving the slide with water, which only slightly reduced the signal. We also tried mixing the SDS solution with the TSMW buffer which we use often in microarray incubation and wash steps, and saw no benefit as compared to SDS alone. Similar results were also obtained when SDS was used with DTT, however due to the noxious odor of DTT, we avoided using it as no apparent benefit was observed. Since our past experience has shown that bleach could be successfully used to isolate glycans from proteins (Song et al. 2016), we applied 0.1% bleach to the slide. While this removed binding, when attempting to reuse the slide we noticed an increase in the background and non-specific binding. As an alternative

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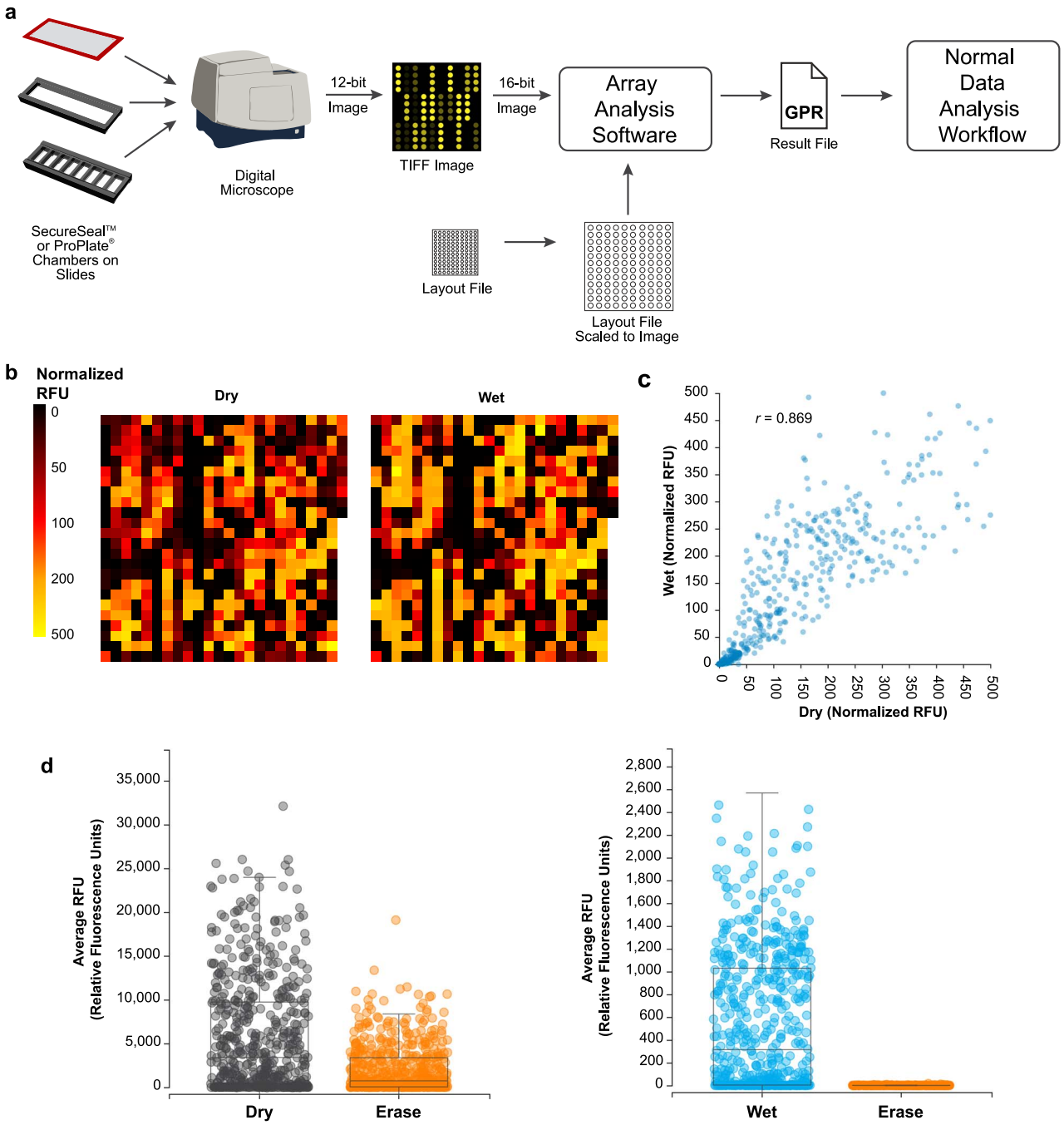


Fig. 1. Establishing the wet scanning and MAWE method. a) The wet scanning workflow shows step by step how an experiment is carried out and the slide scanned under wet conditions. Following the scanning, analysis is performed after making adjustments for the array list files using traditional microarray software. b) Calendar heatmap (map provided in Supplementary Dataset S1, indicating which glycan corresponds to which tile) showing the normalized relative fluorescent units (RFU) values (normalized to 500) for a lectin cocktail applied on the CFG array under dry conditions or wet conditions. Each square represents an individual glycan compound printed on the microarray, and the intensity of binding is represented by the color scale shown alongside. The position of the glycan square in the two blocks represents the same glycan. Both methods show similar binding patterns and the "Wet" assay shows a tendency to bind relatively more. c) Scatterplot of the data where each point represents a glycan, and the x and y position of the spot is determined by normalized RFU under either wet or dry conditions. Plot is used to calculate Pearson correlation coefficient (r) of 0.869 between the two methods. d) Jitter plot where each glycan is represented by data point under either dry or wet conditions, followed by MAWE technique to erase the binding. The dry method shows a signal decrease of ~40%, while the wet method shows a decrease of >95% establishing the need for the slide to remain wet at all times for efficient cleaning.

approach, we tried hapten inhibition to GBPs, using a series of conditions with a mix of monosaccharides in order to try to compete with the binding. The mix contained glucose, galactose, mannose, fucose, N-acetylglucosamine (GlcNAc), N-acetylgalactosamine (GalNAc) and sialic acid at 50 mM. The mix was applied to wash the slide using quick washes, long incubations or microwaves. Yet, these conditions did not result in sufficient reduction of binding.

In conclusion, washes with 10% SDS with short times of microwaving gave the best outcomes for cleaning the slide and appeared to have no effect on GBP binding in subsequent studies. We tried to clean the slide comparing dry and wet

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method with the 10% SDS and microwave. As can be seen, the maximum signal for the wet method drops by >95%, but the signal for the slides which were dried for scanning drops by only 40% (Fig. 1d). This supports our hypothesis that in order to be able to clean and reuse the slides, they should be kept hydrated at all times.

Reusability of CFG Array with biotinylated lectin cocktail

To test whether this method enables reusability, we tested a single CFG array using the same lectin cocktail (ConA, GNL, PHA-E, PHA-L, MAL-I, SNA, AAL, LCA, RCA-I, GSL-I, GSL-II, each at 2 μg/mL) multiple times with streptavidin-Cy5 at 0.5 μg/mL as the detection reagent. Between each assay run, the microarray was cleaned using MAWE method (Fig. 2a), before subsequent reuse. In certain cases, some binding was resistant (e.g. between run #5 and #6) and eventually required multiple erase steps for complete removal of binding. As can be seen, the overall pattern of binding of the lectin cocktail did not change appreciably over the course of 15 runs of assays. In fact, the similarity of binding was quite remarkable between run #1 and run #15 with a Pearson correlation coefficient of 0.985 (Fig. 2b). The average RFU of binding between the runs was for the most part consistent throughout the 15 runs (Fig. 2c), except for run #3 and #5. In these two runs, we had used a lectin cocktail sample prepared on the previous day, and this was likely the reason why we observed lower overall binding. Fresh lectin preparations were used for all other runs and showed consistent binding.

Reusability of NCFG Array with biotinylated lectin cocktail

Since the NCFG arrays were printed in a multi-array format (8 arrays/slide), we wanted to check if the same reusability could be achieved in this system. We applied the same cocktail of lectins multiple times with streptavidin-Cy5 as the detection reagent and similar to the CFG array, the NCFG array was cleaned between assays using the erase method (Fig. 3a). During the first few runs, we had to optimize the microwave settings and times for the smaller array and multi-well nature of these arrays, which required multiple erase steps. Once optimized, we used 10x washes with 10% SDS in microwave at high settings for 10 s to perform the erase steps. However, due to the smaller nature of the NCFG arrays, it is possible that in some cases the array being cleaned was not evenly microwaved due to formation of hot-spots in the microwave, and eventually, some later runs required more washes. Yet, the overall binding pattern between the runs remained quite similar and the Pearson correlation coefficient between run #1 and run #15 was 0.942 (Fig. 3b). The average RFU also remained consistent throughout the 15 runs (Fig. 3c). Note: We observed that run #2 showed lower correlation to other runs, as our records indicated that we used lectin cocktail from the previous day, as mentioned above, and the same issue was seen in run #5. In all other runs, freshly prepared lectin cocktail was used.

Sequential application of biotinylated lectins on NCFG Array

To test if MAWE shows consistent results with different samples, we applied 3 lectins (ConA, AAL and RCA-I) to the NCFG array sequentially, rather than as a cocktail. Each lectin was applied individually over 6 runs and detected with streptavidin-Cy5. Each lectin was tested twice overall, with alternating lectins in between the sequence of ConA, AAL, RCA-I, followed by another round of ConA, AAL, and RCA-I (Fig. 4a). The data shows that the slide could be quantitatively cleaned after each array experiment and could be used for subsequent lectin analysis. Furthermore, the data indicate that the same lectin, when applied after several rounds of reuse, still exhibits highly correlated data to the first set of runs, with a Pearson correlation coefficient ranging from 0.991–0.997 (Fig. 4b).

In attempting this set of experiments with a larger set of lectins, we discovered that the lectin LEL, appeared resistant to cleaning using the MAWE method (Supplementary Fig. S2a). We attribute this to the highly glycosylated nature of LEL (~50% glycan by weight) which is known to have highly unusual hydroxyprolines containing galacto-arabinose polymers (Nachbar et al. 1980; Peumans et al. 2003). To verify that this property of the sample was the issue, we tested another similarly highly glycosylated lectin, STL, which showed similar properties of resistance to this cleaning method (Supplementary Fig. S2b). Like LEL, STL is also known to contain hydroxyprolines, with heavy glycosylation (Allen et al. 1996). These results therefore suggest that such heavily glycosylated proteins could be resistant to denaturation, and resistant to the standard MAWE method of erase; such proteins could require additional wash steps.

Reusing NCFG Arrays to assess serum from healthy individuals

Glycan microarrays have been useful in extensively characterizing the anti-carbohydrate antibody repertoires of human serum (von Gunten et al. 2009; Muthana and Gildersleeve 2014; Schneider et al. 2015; Durbin et al. 2018; Jandus et al. 2019; Leviatan Ben-Arye et al. 2019; Luetscher et al. 2020). To show that the MAWE method can be used with serum, we tested the IgG binding of serum from anonymous healthy individuals. Our initial attempts to use fluorophore conjugated anti-IgG antibodies, such as Alexa 633- or Cy5-conjugated anti-human IgG resulted in very low signal. We realized that this lower signal was a result of quenching due to the solvent and environmental effects on these fluorophores when conjugated to antibodies and present in wet conditions in buffer, which is a well-known phenomenon (Lakowicz 2006). However, when these fluorophores were conjugated to streptavidin (like Cy5), the protein seemed to shield it from environmental effects, resulting in good quantum yield. We therefore strategized to use biotinylated anti-human IgG with streptavidin-Cy5 which resulted in ~10 times increase in signal.

Once we optimized the detection method, we tested serum samples on the NCFG microarray for two individuals with ID# 8 and ID# 9 (Fig. 5a), two times each in random order, with MAWE method in between to remove any binding. We observed binding on the array for these individuals, which showed binding or lack of binding to specific blood group glycans, which can be useful in characterizing their anti-blood type repertoire. This profiling of anti-blood group antibodies using microarrays can be very useful in transfusion medicine and also in pathological conditions (Lee-Sundlov et al. 2020; Luetscher et al. 2020; Kim et al. 2021). We observed that the MAWE method produced very consistent results for serum as well, with intra-sample Pearson correlation coefficients (r) of 0.951 and 0.989, showing good reproducibility (Fig. 5b).

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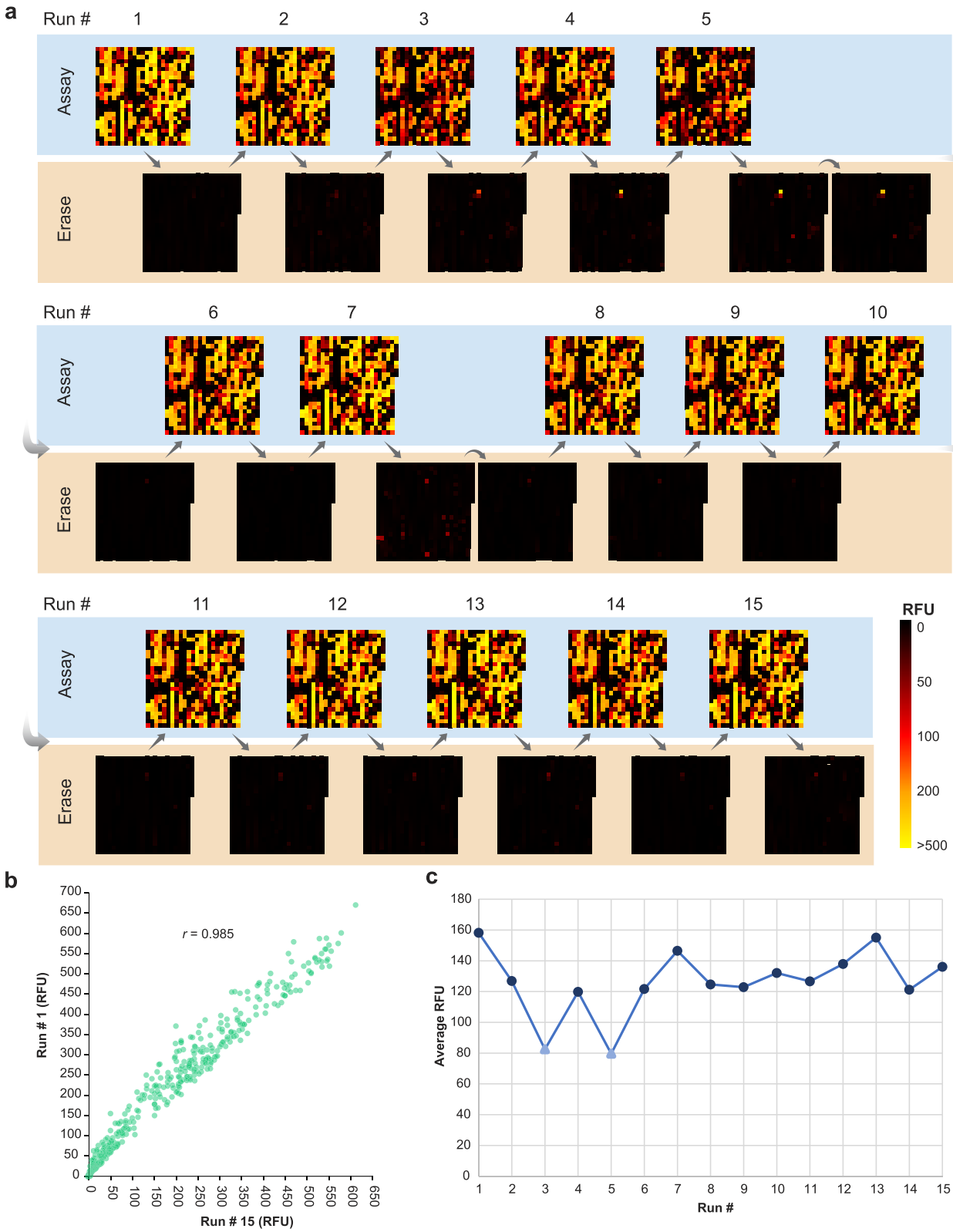


Fig. 2. Reusability of CFG microarrays. a) Calendar heatmaps (map provided in Supplementary Dataset S1, indicating which glycan corresponds to which tile) of unnormalized binding on the CFG array. Each block represents data for either an assay performed by application of lectin cocktail, or the corresponding erase step performed using MAWE to clean the slide for subsequent use. A single slide could be cleaned and reused 15 times. Certain glycans were more resistant to cleaning and needed more rounds of cleaning (e.g. between run #5 and #6), but did not affect subsequent assay results. Binding pattern between all 15 runs are very similar as can be seen. b) Scatterplot of the data where each point represents a glycan, and the x and y position of the spot determined by RFU value between run #1 and run #15. Plot shows a Pearson correlation coefficient (*r*) of 0.985, showing the high reproducibility of the method over 15 rounds of assays on a single slide. c) Average RFU for each assay run plotted vs the run #, shows relatively unchanged overall binding. Run #3 and #5 (indicated by triangle makers) shows slight dip in the average RFU values as lectin cocktail from the previous day was used.

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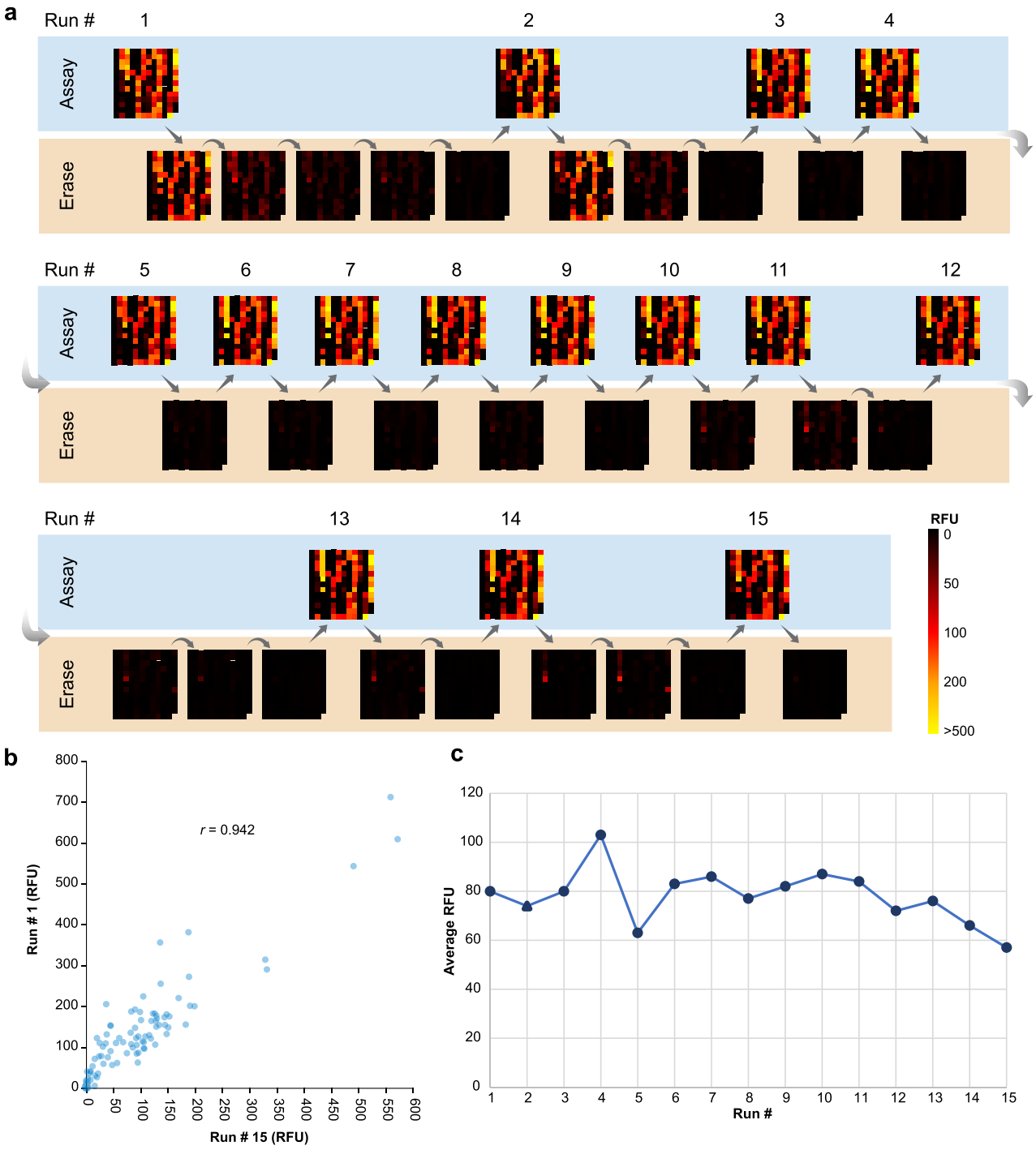


Fig. 3. Reusability of NCFG microarrays. a) Calendar heatmaps (map provided in Supplementary Dataset S2, indicating which glycan corresponds to which tile) of unnormalized binding on the NCFG array. Each block represents data for either an assay performed by application of lectin cocktail, or the corresponding erase step performed using MAWE to clean the slide for subsequent use. A single NCFG array could be cleaned and reused 15 times. During the first run (between run #1 and run #2), we had to optimize the microwave conditions for the smaller array, which forced us to clean multiple times. Certain glycans were more resistant to cleaning and needed more rounds of cleaning (e.g. between run #12 and #13), but did not affect subsequent assay results. Binding patterns between all 15 runs were very similar. b) Scatterplot of the data where each point represents a glycan, and the x and y position of the spot determined by RFU value between run #1 and run #15. Plot shows a Pearson correlation coefficient (r) of 0.942, showing the high reproducibility of the method over 15 rounds of assays on a single slide. c) Average RFU for each assay run plotted vs the run #, shows relatively unchanged overall binding. Run #2 (indicated by triangle makers) shows slight dip in the average RFU values as lectin cocktail from the previous day was used.

Live cell binding on glycan microarrays

One advantage of using a digital microscope is that it can be used to view binding of live cells to glycan microarrays. As a proof of concept, we used lamprey Variable Lymphocyte Receptors (VLRs) cloned into yeast surface display (YSD) library, which we described previously (McKitrick et al. 2020).

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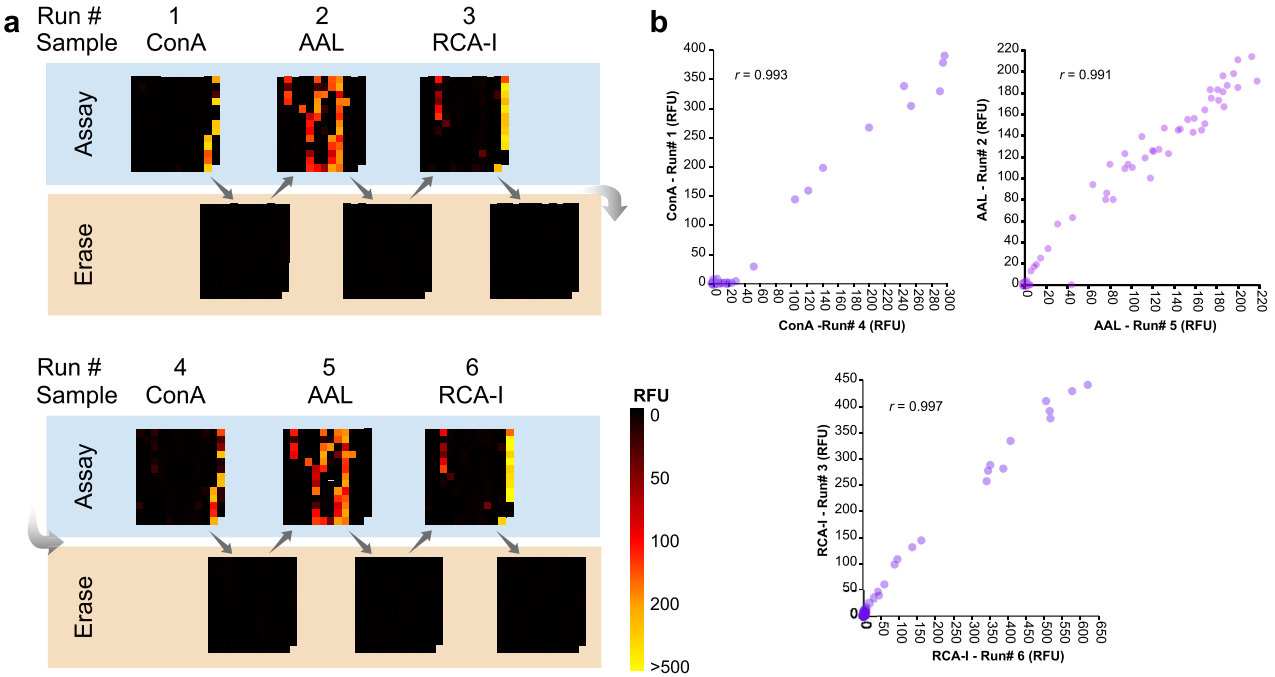


Fig. 4. Sequential application of lectins on NCFG array. a) Calendar heatmaps (similar to Fig. 3) showing binding on the array for either ConA, AAL or RCA-I lectins. The same lectin panel repeated two times for a total of 6 runs sequentially, with cleaning in between each run. Binding patterns for the same lectin after other lectins in between remain largely similar. b) Scatterplots of the data where each point represents a glycan, and the x and y position of the spot was determined by RFU value between the first round of the lectin and the corresponding next round of the same lectin. The data shows high correlation between the two rounds of the same lectin despite being run with different samples in between. Pearson correlation coefficient of ConA run #1 vs ConA run #4 = 0.993, Pearson correlation coefficient between AAL run #2 vs AAL run #5 = 0.991, Pearson correlation coefficient between RCA-I run #3 vs RCA-I run #6 = 0.997.

We expressed two of these VLRs, namely the O6 and the Tn4-11, in two sets of yeast. The expression of these VLRs on the surface was confirmed using flow cytometry (Supplementary Fig. S3a). The O6 expressing yeast was incubated on the CFG microarray and exhibited binding to the expected glycans on the surface (Fig. 6a and b, Supplementary Fig. S3b). We were able to pick the yeast from the slide, plate them on SDCAA plates, and found that the yeast were still viable for growth in medium (Supplementary Fig. S3c).

The yeast were then erased from the slide with MAWE protocol and next the Tn4-11 yeast was incubated on the microarray. The Tn4-11 yeast also showed binding on the microarray similar to what was previously published (Fig. 6a and b), after which it was also erased from the slide using the MAWE method (Fig. 6a). These results confirm that the MAWE method and wet scanning have potential for screening whole live cells on glycan microarrays in a reusable manner.

Discussion

Glycan microarrays are a valuable and informative resource to study glycan-protein interactions, but current technologies limit arrays to a single analysis. This limitation coupled with the restricted availability of the glycan reagents creates hurdles for expanding studies of glycan-protein interactions. Our findings here present a new method of assaying and scanning slides that have been maintained under "wet" conditions, and further demonstrate a robust method of cleaning the slides using MAWE for reuse. We have shown that this method can provide reproducible results with a cocktail of biotinylated lectins on two types of arrays (i.e. the CFG and NCFG arrays), at least 15 times. The robustness of the technique is further validated by applying lectins in a sequential manner. We also show that this technique can be used for a variety of biologically relevant sample types including human sera and whole live cells such as yeast expressing specific lamprey VLRs.

The key to this method lies in keeping the slides hydrated at all times, and using SDS with microwaves to clean the slides of protein reagents between uses. SDS is known to have minimal adsorption on glycans, but to bind tightly to proteins and can induce denaturation (Bhuyan 2010). Previous studies have shown that SDS interacts with glycans almost 3–10 times lower than peptides, making it a good denaturant of the bound proteins without causing any detrimental effect on the glycans printed (Bagger et al. 2007). However, this property of SDS is also the reason why certain heavily glycosylated proteins (such as LEL and STL lectins) might be resistant to the erase steps as compared to others. The application of microwaves is also essential for the process as hot SDS alone does not affect removal of bound proteins.

We propose the following mechanism to explain the MAWE approach (Fig. 7). The surface of the glycan microarrays is a 3D polymer matrix likely to be many nm in thickness in the dry condition, as seen for polyethylene-based approaches (Harbers et al. 2007), which in the wet condition could entangle the proteins binding to them. After being dried, the polymer matrix may collapse around the protein, thus permanently embedding the protein into the surface. Under wet conditions, however, this polymer matrix remains expanded, allowing for penetration of SDS; polymer entanglement, however, with the protein may still remain. Microwaving the slide excites water molecules between the polymer and the protein,

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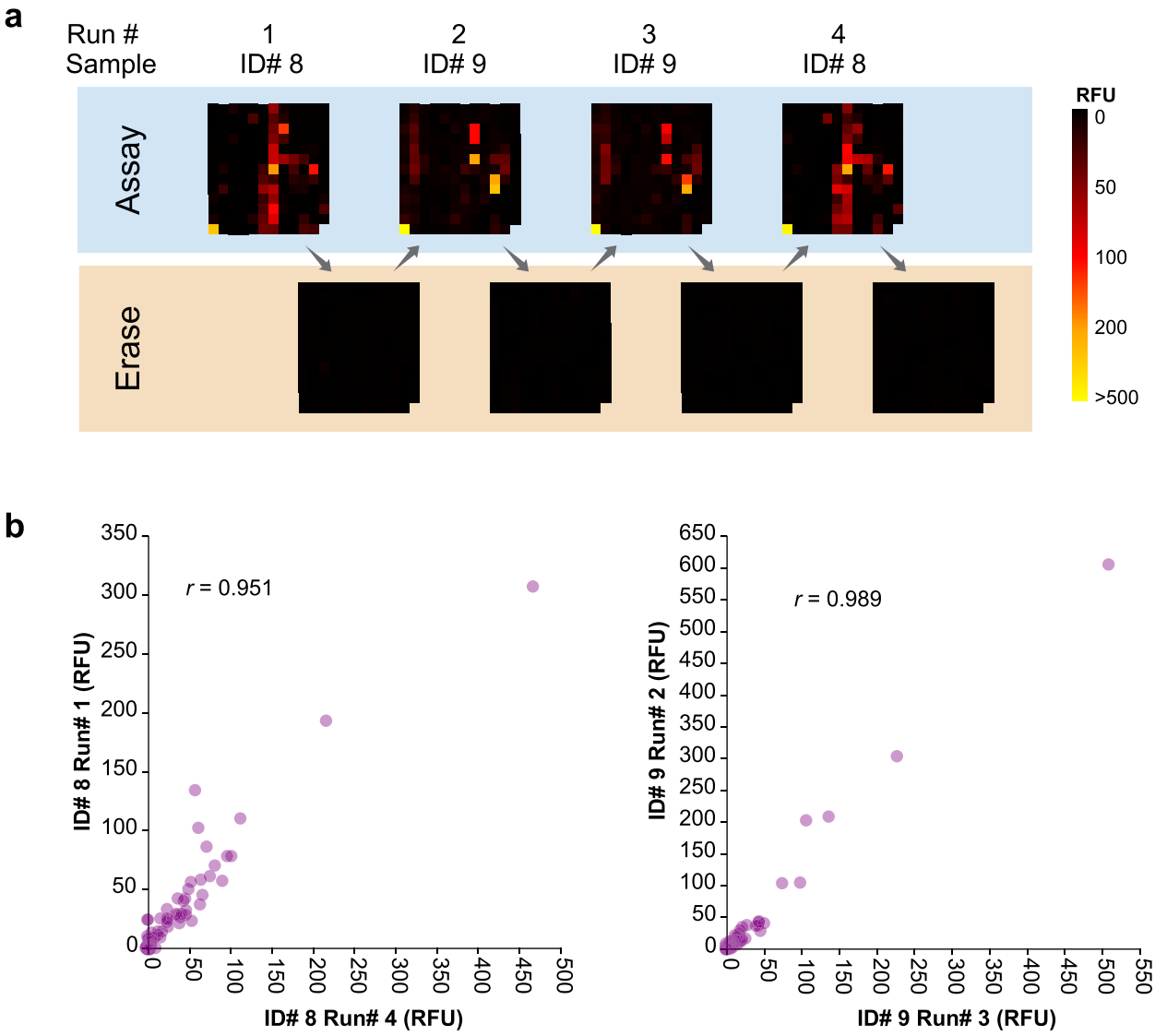


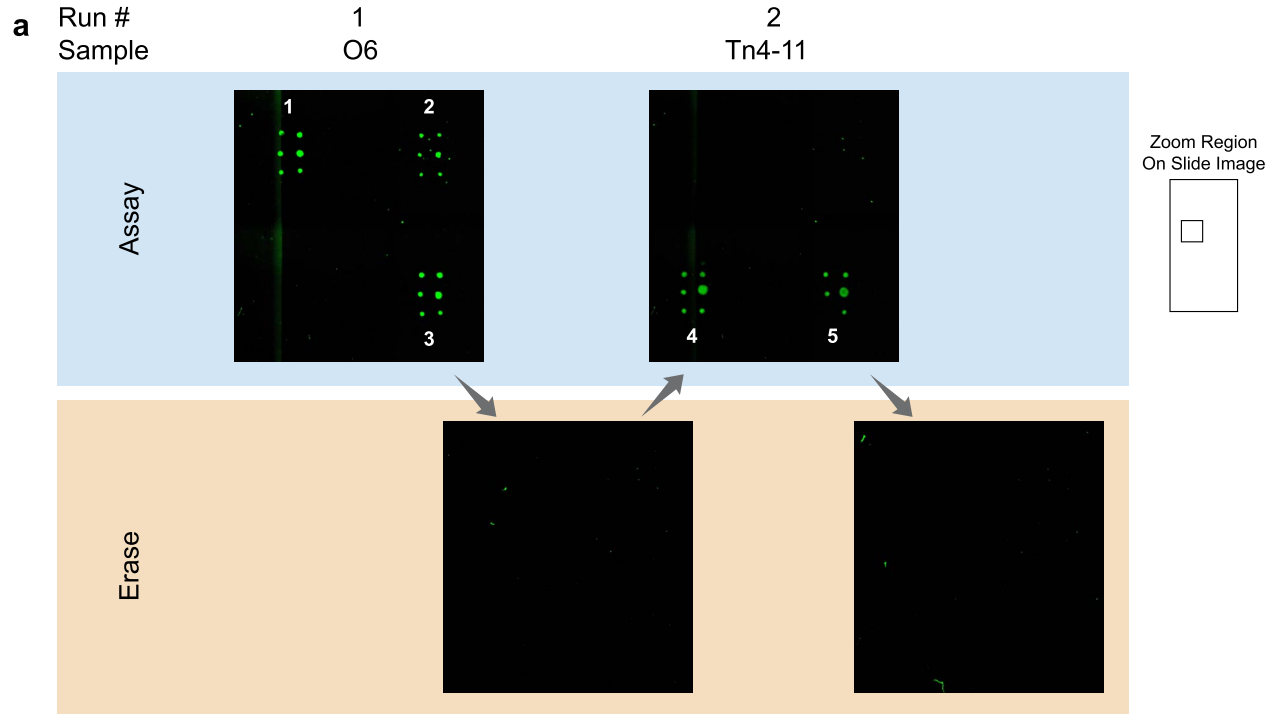
Fig. 5. Healthy human serum on an NCFG array. a) Calendar heatmaps (similar to Fig. 3) showing binding on the NCFG array for serum from healthy individuals with ID# 8 or 9. Individual with ID# 8, predicted to be an O blood group, shows antibodies against both A and B blood group glycans, while that of ID# 9, predicted to be an AB blood group, lacks antibodies against A and B glycans. Between each serum assay, the slide was cleaned using the MAWE erase method which showed complete removal of all binding. b) Scatterplots of the data where each point represents a glycan, and the x and y position of the spot determined by RFU value between the first round of the individuals with ID# 8 (left panel) or ID# 9 (right panel) and the corresponding next round of the same individual. As can be seen, the data shows that the method is very reproducible, showing Pearson correlation coefficients of 0.951 (ID# 8, runs 1 and 4) and 0.989 (ID# 9, runs 2 and 3).

to further expand and mobilize the polymer, and promotes SDS mobility to reach the bound protein to denature it, resulting in its release from the glycan ligand and allowing the protein to be cleaned off the slide. After removal of the SDS solution, and maintaining the wet condition, the slide can be either stored or reused with the next sample.

The MAWE method to allow slide reusability should extend the use of this limited resource many-fold, and also provides a method to allow related samples to be tested on the same printed surface. The use of a single microarray to test multiple samples also provides additional quality control for the experiment, as at any point in the sets of experiments, one can retest the sample again to verify repeatability. Furthermore, one could add a known GBP (e.g. lectin) to quality control an array, at any point in its reuse, before putting an unknown sample on the same array—thereby adding a validation of the specific array being used and even to a specific type of glycan. Such sequential testing is not available on conventional dry array methods.

A limitation of the wet scanning is that the choice of fluorophores creates a restriction compared to those which are not affected by solvent and environmental effects. However, such a limitation can be overcome, as we have shown, in one example by using biotinylated detection reagents with streptavidin-Cy5. Another limitation is that when using single well ProPlate or the SecureSeal, at least 1 mL of sample has to be applied, which is greater than the typical amount of 100 μL of sample required for the coverslip technique. We feel encouraged that in the future this issue will be addressed by using custom-designed chambers which can hold lesser volumes in a robust manner. A limitation with the instrument which we used is that due to the LED-based

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b

	O6 Binders			Tn4-11 Binders	
	3S β4 6S β Sp8 **1**	3S β4 β Sp0 **2**	3S β4 6S β Sp0 **3**	α Sp8 **4**	α Sp11 **5**
Rank - Yeast VLR	3	2	1	1	2
Rank - VLR-Fc*	1	4	2	1	2

* Data from McKitrick, T.R., Goth, C.K., et al. 2020

Fig. 6. Live yeast expressing monoclonal VLR through YSD bind on the array preferentially. a) Zoomed in images from the digital microscope showing two types of yeast, expressing either O6 or Tn4-11 VLRs (as described previously (McKitrick et al. 2020)) on their surface incubated on the CFG array sequentially, with MAWE protocol in between to erase and reuse the slide. The numbers in white next to the spots correspond to the structure number shown in panel b. b) Structures of the 5 glycans which show binding in panel a and the rank of the glycan binding (out of >550 glycans) on the array for the yeast displaying the VLR as compared to the VLR-Fc chimera published previously (McKitrick et al. 2020). Full binding data presented in Supplementary Fig. S3b. Glycan structures drawn using GlycoGlyph at https://glycotoolkit.com/glycoglyph (Mehta and Cummings 2020).

illumination in the ImageXpress Pico (wet), the overall signal intensities were lower, yet the patterns of the signals were still very comparable to the GenePix (dry) scanner. This issue could be resolved in the future by using a laser-based microscope, which should provide more sensitivity. We noticed that certain proteins were resistant to cleaning using our method. Future studies could develop cleaning solutions which may work better for such resistant proteins. Despite these limitations, there are still many obvious benefits of being able to reuse glycan array slides.

Previously, live cells have been imaged in lectin microarrays (Li et al. 2011), but the number of such studies is relatively meager. However, this study used immobilized lectins and was used for interrogating glycans on the surface of the cells, rather than identifying which glycans the cells can bind, per a typical glycan microarray approach. Another study described the use of glycan microarrays for binding porcine sperm (Kadirvel et al. 2012). In this case, the slide was placed glycan side down on a suspension of uncapacitated sperm, so that only motile cells could swim up in the medium and bind to the glycans on the slide. A more recent study showed how one could recombinantly express a protein Chaperone-usher fimbriae in *E. coli* and use glycan arrays to characterize the binding (Day et al. 2021). Another study described binding of *E. coli*

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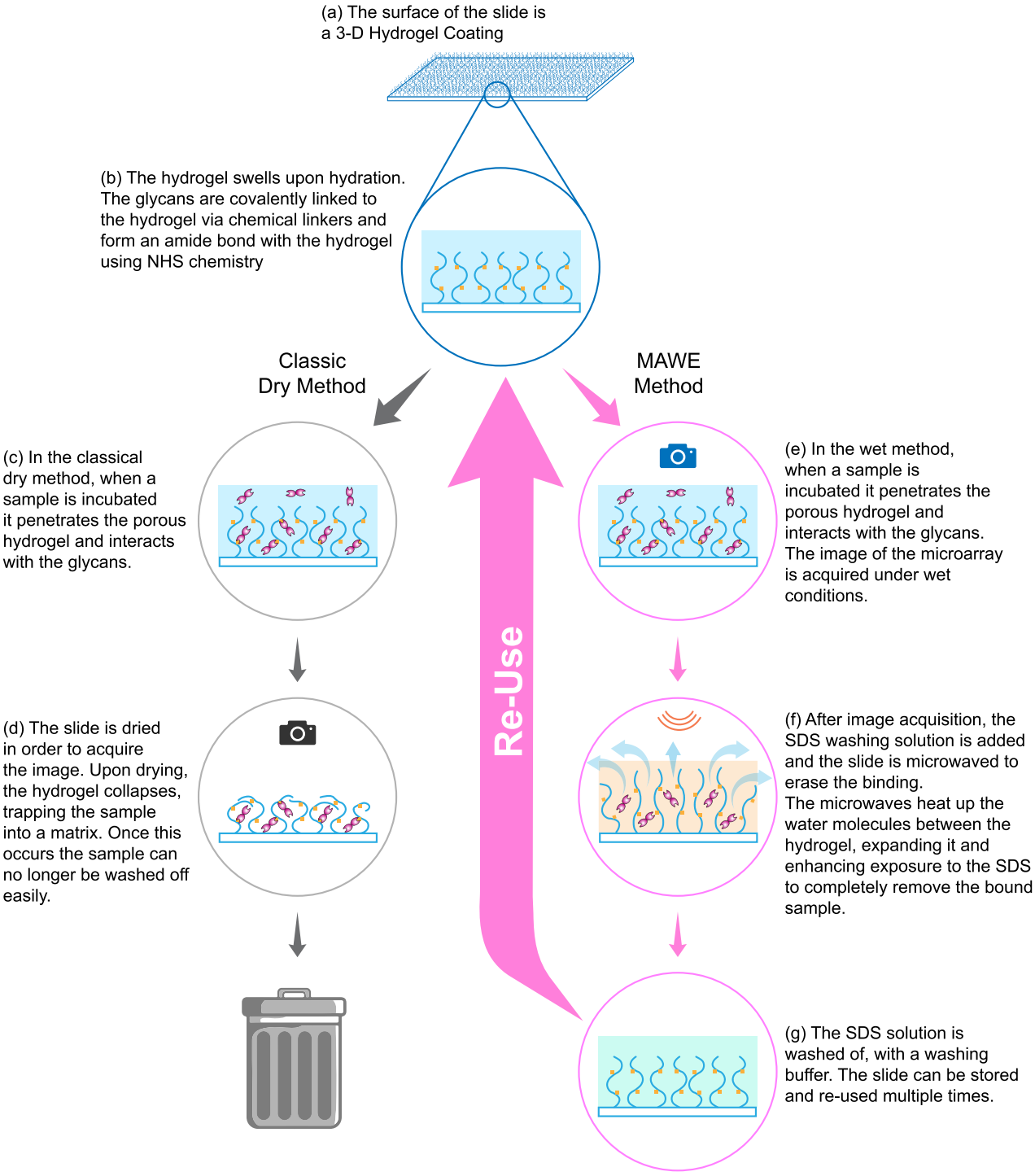


Fig. 7. Schematic comparison of the dry and MAWE methods. a and b) The surface of the microarray slide contains 3-D hydrogel coating which is invisible to the naked eye. This hydrogel contains the glycans (small yellow squares) covalently attached via linkers using NHS chemistry. When the slide is hydrated, this hydrogel surface swells. (c and d) In the classical dry method, the sample is incubated on the wet hydrogel surface. The sample penetrates the porous hydrogel and interacts with the glycans. The slide is then dried, which causes the hydrogel to collapse and traps the sample. The image is acquired, and once the hydrogel collapses onto the protein/sample, it can no longer be washed off. (e–g) In the wet method described here, the sample is incubated on the slide and interacts with the glycans. After application of the secondary detection reagents the image is acquired while maintaining these wet conditions. After acquisition, the solution on the slide is replaced with SDS solution, and is microwaved repeatedly to remove the binding. We hypothesize that the microwaves excite the water molecules within the hydrogel, thereby causing the hydrogel to expand further and increase exposure of the bound proteins/sample to the SDS, thus assisting with the erase. Once the sample is erased, the SDS solution is washed off and replaced with either preservative buffer to store overnight or can be reused directly for the next experiment.

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and *Helicobacter pylori* to multivalent glycan microarrays (Kim et al. 2018). However, in this study the binding was not quantified, and was only binarily categorized as qualified yes/no binding. In all of these studies, it was not mentioned whether the slides could be reused. Nevertheless, such studies illustrate how important biological functions can be studied using glycan microarrays. Our yeast experiments demonstrate that living yeast can bind glycans on the microarrays, and that the arrays are reusable using our method of wet scanning. Moreover, yeast binding patterns were similar to those previously published for the purified proteins which they express. The ability to reuse the slides for multiple such cells expands the utility of this important screening platform to identify glycan binding organisms, and creates new possibilities for on-slide binding assays with a multitude of different types of samples and conditions.

In conclusion, these studies represent the first demonstration of robust reusability of glycan microarrays. The MAWE technique shown here should be useful in conserving glycan microarrays, and the precious glycan reagents needed to generate them, while at the same time enabling more robust and complex studies on GBPs, antibodies, and cells.

Materials and methods

Materials

Overlays (either SecureSeal® or ProPlate® Hybridization Chambers) were added to the slide surface to allow for the addition and removal of reagents. SecureSeal® Hybridization Chambers (cat.# 621506), ProPlate® 1-Well Slide Module with Delrin Snap Clips were used with custom silicone gasket to fit the CFG arrays, ProPlate® 8-Well Slide Module with Delrin Snap Clips (cat.# 246868), and ProPlate® 24-Well Slide Module with Delrin Snap Clips (cat.# 472757) all from Grace Bio-Labs were used for this purpose. When necessary, the overlays were sealed with Adhesive PCR Plate Seals (cat.# AB0558) from ThermoFisher Scientific, to minimize evaporation. TSM buffer and derivatives of this buffer were used throughout the assays. TSM buffer had a composition of 20 mM Tris-HCl, 150 mM NaCl, 2 mM $CaCl_2$, 2 mM $MgCl_2$, with a pH of 7.4. TSM Wash Buffer was made by adding 0.05% Tween-20 to the TSM buffer. TSM Binding Buffer consisted of TSM buffer plus 0.05% Tween-20 and 1% BSA-protease free. TSM Preservative Buffer was made by adding 0.05% Tween-20, 0.02% Sodium Azide, and 5% Glycerol to TSM buffer. All biotinylated lectins were obtained from Vector Laboratories and have been previously tested for specificity within the NCFG and in collaboration with Vector Laboratories (https://research.bidmc.org/ncfg/data-ncfg/microarray-data/lectin-quality-assurancequality-control).

All serum samples were obtained through the Ragon Institute Healthy Control Cohort Study comprised of healthy 18–65-year-olds in the Boston area. The Massachusetts General Hospital Institutional Review Boards (IRB) approved the study (Luetscher et al. 2020). EBY100 yeast clones expressing monoclonal Tn4-11 (McKitrick et al. 2020) and O6 (McKitrick et al. 2021) VLR proteins as Aga2P-anchored yeast-surface display were generated as previously described (McKitrick et al. 2022) and cryopreserved as glycerol stocks. Various secondary detection reagents were used based on the sample. These regents included biotinylated-anti-human IgG (cat.# 109-065-088) from Jackson ImmunoResearch, streptavidin-Cy5 (cat.# 434316) from Invitrogen, and anti-myc tag (cat.# 3693896) from Millipore. Images were captured either on the ImageXpress Pico imager or Genepix 4400A scanner (both from Molecular Devices). They were then analyzed with Genepix Pro 7 Microarray Acquisition and Analysis Software by Molecular Devices and subsequently viewed in GLAD, as described below. Wet erase was performed using a 700 W microwave (Model No. MQS0803W from Quasar) as well as a 10% SDS Solution. To prepare this solution, 10 g of sodium dodecyl sulfate (cat.# L3771-500G) from Sigma-Aldrich was added for every 100 mL of water used. The solution was then sonicated to ensure homogeneity. SDCCA and SGCAA medium, as well as SDCAA plates were prepared as previously described in detail (McKitrick et al. 2022) using reagents from Sigma-Aldrich and Fisher Scientific. Yeast was grown in 100 mm culture tubes (Fisher Scientific, 149569B) and Petri dishes (Fisherbrand, FB0875712).

Description of CFG, NCFG and Lectin QA/QC arrays

In this study, we used two types of microarrays which provided wide diversity in glycans and linkers. The CFG glycan array (version 5.5) consisting of 562 glycan derivatives attached via 1 of 24 linkers and the NCFG glycan array (version 3.2) consisting of 155 glycan derivatives with 2 linkers. The Lectin QA/QC glycan array consists of a subset of 20 glycans derivatives from the NCFG array but provides more arrays/slide (see below). The CFG arrays were obtained from The Scripps Research Institute (Blixt et al. 2004). The NCFG and Lectin QA/QC arrays were printed in-house using the Scienion sciFLEXARRAYER SX as described previously (Luetscher et al. 2020). The CFG glycan array is a single array/slide system enabling the investigation of 1 sample at a time on a slide, while the NCFG glycan array is an 8-array/slide system which enables investigation of 8 samples at a time on a single slide, and the Lectin QA/QC array is a 24-array/slide system which enables the investigation of 24 samples at a time on a single.

Description of data visualizations

Throughout the paper, heatmaps called Calendar Heatmaps are shown, which represent the relative fluorescence units (RFU) as colored squares which are stacked from top to bottom then left to right. These figures are generated using GLAD (Mehta and Cummings 2019). We have provided the map of the chart ID numbers for these Calendar Heatmaps in the Supplementary Dataset S1 and S2.

Application of sample when using SecureSeal

CFG slides used a SecureSeal hybridization chamber or a ProPlate 1 well chamber. The plastic backing was removed from the SecureSeal and the adhesive was placed firmly around the printed area of the slide. A Kim wipe was then used to press on the back side of the slide to make sure the SecureSeal was fully adhered. The SecureSeal Hybridization chamber can hold a volume of 1,000 μL and therefore all solutions were used at a volume of 1,000 μL. When adding samples or secondary detecting agents, no air bubbles were introduced into the SecureSeal solution, as this could prevent binding to the array in that location. If there was an issue of air bubbles occurring, this was remediated by tilting the slide at about a 30° angle vertically with the access port being used to remove and add liquid, down (Supplementary Fig. S4a). Keeping this

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30° angle helped remove and prevent air bubbles when adding and removing solutions, especially solutions with detergent. Occasionally during the microwaving process, the SecureSeal became loose and no longer attached to the slide. When this occurred the SecureSeal was removed from the slide, the slide was dip washed or washed with a squirt bottle with TSM Wash Buffer, TSM Buffer, and water, then dried by centrifugation. A new SecureSeal was then secured to the slide and buffer was immediately added (minimizing the time for which the slide was kept dry). Switching SecureSeals was only done after the MAWE protocol was complete to prevent the slide from drying while the sample was still present. To help prevent the SecureSeal from losing its adhesion to the slide, the slide was lifted by holding only the glass portions of the slide where the SecureSeal was not located.

Application of sample when using ProPlate

NCFG slides used a ProPlate 8-well chamber, CFG slides used a ProPlate 1-well chamber, and Lectin QAQC slides used a Proplate 24-well chamber. The ProPlate was firmly placed around the printed area of the slide and secured into place with Delrin Snap clips. The ProPlate 1 well chamber can hold a maximum volume of 10,000 μL, however all solutions were used at a volume of 1,000 μL, which is sufficient to cover the slide while avoiding spillage and drying. The ProPlate 8 well chambers can hold a volume of 1,000 μL\well however all solutions were used at a volume of 500 μL\well to prevent leakage into adjacent wells. The ProPlate 24 well chambers can hold a maximum volume of 300 μL\well, however all solutions were used at a volume of 150 μL to prevent leakage into adjacent wells. When scanning the NCFG slide with the ImageXpress Pico the slide was not able to fit in the slide holder due to the ProPlate. To remedy this issue a cut was made to remove the top right corner of the Delrin clip touching the bottom of the slide (Supplementary Fig. S4b). The cut was made as small as possible, while still serving its purpose, as to not affect the ability of the chamber to hold liquid without leaking. During the microwaving process the wells not being washed with SDS were covered with adhesive PCR plate seals, cut to fit the size of the ProPate, to prevent splashing of the SDS into nearby wells. Alternatively, all wells could be filled with SDS and cleaned anyways, as we didn't notice any difference.

Setting up focus

To setup the focus of the slide with the ImageXpress Pico, we used a custom printed microarray containing fluorophores printed in a large grid (i.e. Alexa Fluor 488 Hydrazide and Alexa Fluor 633 Hydrazide). The slide was attached with the overlay (either SecureSeal® or ProPlate® hybridization chambers) and the slides were placed face up (i.e. coverslip-up) into the ImageXpress Pico. The slides were focused and the heights offsets for the focus plane was adjusted and stored as separate labware configurations for future use in protocols. This provided a good starting point for fine tuning the focus during the assays.

Scanning and analysis

Using the ImageXpress Pico a stitched slide image was acquired. The scanner was autofocused to both the CFG and NCFG slides and a protocol was created, saving the focus position for each slide. Each image was acquired in a 2.5× objective within an exposure time of 5,000 ms at the appropriate wavelength. After acquisition the image was exported and opened in the GenePix Pro 7 software. The layout file (GAL file) for the corresponding array was opened and aligned with the image. A results file (GPR file) was then generated and converted into an Excel file to be utilized for data analysis. From the Excel file, a Glycan Array Dashboard file (GLAD text file) was created for data visualization using the GLAD software program (https://glycotoolkit.com/GLAD/).

When scanning with the GenePix (dry), a PMT of 450 and Laser Power of 70 were used. Once the image was acquired it was processed in the GenePix Pro 7 software similar to above.

Wet vs dry scanning concentration sensitivity assay

The slide was removed from the freezer and immediately placed in a vacuum desiccator for 20 min. The appropriate overlay was then secured to the slide, as described above. The slides were rehydrated for 5 min with TSM Wash Buffer. The AAL biotinylated lectin sample was diluted to the desired concentrations in TSM Binding Buffer. Ten concentrations were used ranging from 20 μg/mL to 0.02 μg/mL (20 μg/mL, 10 μg/mL, 5 μg/mL, 2 μg/mL, 1 μg/mL, 0.5 μg/mL, 0.2 μg/mL, 0.1 μg/mL, 0.05 μg/mL, 0.02 μg/mL). TSM Wash Buffer was removed from the slides and replaced with the same volume of sample which incubated on a shaker at RT for one hour. The slides were then washed four times with both TSM Wash Buffer and TSM Buffer. Fluorescent Streptavidin Cy5 (SA-Cy5) was then diluted to a concentration of 0.5 μg/mL in TSM Binding Buffer, and this was added to the slides and incubated for one hour shaking at RT. The slides were then washed four times with both TSM Wash Buffer and TSM Buffer and then kept hydrated with TSM Buffer. The slide was scanned using the ImageXpress Pico as described above. The slides were then immediately washed four times with water and aspirated to dry. The slide was scanned using the GenePix scanner as described above.

Microwave assisted wet-erase process

The slide was washed two times with 10% SDS to replace any existing buffer, then 10% SDS was added to the slides. The slides were then individually placed in a glass container and microwaved. CFG slides were microwaved on a low power setting for 3 s. Note- the time and power used depends on various factors (e.g. microwave power, overlay type and materials, ability of microwave to distribute the energy evenly etc.), so this may need to be optimized for other microwave machines. The CFG slide was then removed from the glass container, the remaining SDS solution was aspirated from the slide, and new 10% SDS was added to the slide. The slide then underwent another round of microwaving. This process was repeated for a total of 10 rounds of microwaving with the SDS being replaced between each round. NCFG slides underwent the same process of 10 rounds of microwaving, however they were microwaved on a high-power setting for 20 s. Both slides were then washed two times with 10% SDS and 4 times with both TSM Wash Buffer and TSM Buffer. The slides were then kept hydrated with TSM Buffer and scanned to verify binding was removed. Binding was classified to be removed if all RFU values were below 30 RFU. If binding was still present, this process was repeated to remove the bound material. It was made sure that the slide went through this washing procedure as soon as the images were taken, in order to ensure an easier sample removal.

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Subsequent assays

Between assays, the slides were kept hydrated with TSM Preservative Buffer and in a humidifying chamber at 4 °C. The humidifying chambers were made by placing a damp Kim Wipe in a Petri dish. When subsequent assays were performed the same protocol was followed. However, since the slide was already hydrated, the rehydration step was replaced with washing the slide twice with TSMW prior to adding the sample.

In our hands, we have managed to reuse a slide for up to 6 weeks after dormant storage (i.e. no usage in between) and up to 3 months when actively used (assays being performed every other day), without change in binding patterns. However, the reusability of the slide could be very sample dependent and therefore it is important to run regular quality controls (e.g. lectin cocktails and secondary only controls) in between samples. We have tried to dry the slides in between runs to see if we could prolong usage, but in our experience, this caused a decrease of signal and increase in background over several reuses. We therefore recommend keeping the slide in the storage buffers rather than drying in between.

Biotinylated lectins on glycan microarray assay

The slide was removed from the freezer and immediately placed in a vacuum desiccator for 20 min. The appropriate overlay was then secured to the slide, as described above. The slides were rehydrated for 5 min with TSM Wash Buffer. All biotinylated lectin samples were diluted to the desired concentrations in TSM Binding Buffer. Typically, a concentration was used between 2 μg/mL and 20 μg/mL. Lectin cocktails were used at a concentration of 2 μg/mL for each lectin while individual sequential lectins were used at 20 μg/mL. TSM Wash Buffer was removed from the slides and replaced with the same volume of sample which incubated on a shaker at RT for one hour. The slides were then washed four times with both TSM Wash Buffer and TSM Buffer. Fluorescent streptavidin-Cy5 (SA-Cy5) was diluted to a concentration of 0.5 μg/mL in TSM Binding Buffer, and this was added to the slides and incubated for one hour shaking at RT. The slides were then washed four times with both TSM Wash Buffer and TSM Buffer and kept hydrated with TSM Buffer. The slides were then scanned using the ImageXpress Pico. The slides were wet-erased using MAWE, washed and stored for subsequent assays following the steps as described above.

Human serum on glycan microarray assay

The slide was removed from the freezer and immediately placed in a vacuum desiccator for 20 min. The appropriate overlay was then secured to the slide. The slides were rehydrated for 5 min with TSM Wash Buffer. All human serum samples were diluted with a 1:20 dilution in TSM Binding Buffer. TSM Wash Buffer was removed from the slides and replaced with the same volume of sample which incubated on a shaker at RT for one hour. The slides were then washed four times with both TSM Wash Buffer and TSM Buffer. Biotinylated anti-human IgG was then diluted to a concentration of 5 μg/mL in TSM Binding Buffer, and this was added to the slides and incubated for one hour shaking at RT. The slides were then washed four times with both TSM Wash Buffer and TSM Buffer. Fluorescent streptavidin-Cy5 (SA-Cy5) was then diluted to a concentration of 0.5 μg/mL in TSM Binding Buffer, and this was added to the slides and incubated for one hour shaking at RT. The slides were then washed four times with both TSM Wash Buffer and TSM Buffer and then kept hydrated with TSM Buffer. The slides were then scanned using the ImageXpress Pico. The slides were then wet-erased using MAWE, washed and stored for subsequent assays following the steps as described above.

Preparation of yeast expressing monoclonal VLRs

Cryopreserved O6 and Tn4-11 monoclonal yeast was thawed, pelleted at 1,000 × g for 5 min, washed once using SDCAA medium, pelleted again and then grown overnight in SDCAA medium at 30 °C and 225 rpm shaking. The next day, yeast was pelleted and VLR surface expression was induced for at least 24 h at 20 °C and 225 rpm. To assess the surface expression of VLRs, 100 μL of the yeast culture was pelleted, stained for 30 min using Alexa Fluor 488-conjugated anti-myc antibody (0.5 μg/mL) in PBS + 0.05% Tween (PBS-T), washed twice using PBS-T and then acquired on an Attune NxT flow cytometer (Invitrogen). Flow cytometric data was analyzed and visualized using FlowJo Version 10.0 (FlowJo LLC).

Yeast on glycan microarray assay

The CFG slide was removed from the freezer and immediately placed in a vacuum desiccator for 20 min. Only the ProPlate overlays were used when working with yeast to prevent sticking of the yeast to the top of the SecureSeal overlays. The slides were rehydrated for 5 min with TSM Wash Buffer. All yeast samples were diluted to a concentration of 100 million cells/mL in TSM Binding Buffer. TSM Wash Buffer was removed from the slides and replaced with the same volume of sample, ProPlate was covered with adhesive PCR plate seal and then incubated on a rocker at 4 °C for at least 3 h. Slides were carefully washed three times with TSM Wash Buffer so as to not disturb the yeast bound to the array. When performing the yeast assays it was made sure that all steps were carried out in the presence of Tween in order to prevent disturbing the yeast during washes. To detect array-bound yeast, slides were incubated under shaking with Alexa Fluor 488-conjugated anti-myc tag antibody diluted to a concentration of 0.5 μg/mL in TSM Binding Buffer for one hour at RT. The slides were then washed three times with TSM Wash Buffer and kept hydrated with TSM Wash Buffer until they were scanned using the ImageXpress Pico scanner. Background images were captured between assays and averaged. This background image was subtracted from the acquired images using ImageJ (FIJI) (version 1.53t), which was subsequently used for the quantification using GenePix Pro 7 and presented in Fig. 6a and Supplementary Fig. S3b. To recover the slides for subsequent assays, they were wet-erased using MAWE and stored following the steps described above.

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Supplementary material

Supplementary material is available at *Glycobiology Journal* online.

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Akul Mehta (Conceptualization [equal], Data curation [lead], Formal analysis [lead], Investigation [equal], Methodology [equal], Visualization [equal], Writing—original draft [lead]), Catherine A. Tilton (Conceptualization [supporting], Data curation [supporting], Formal analysis [equal], Investigation [equal], Methodology [equal], Visualization [equal], Writing—original draft [supporting]), Lukas Muerner (Conceptualization [supporting], Formal analysis [supporting], Investigation [supporting], Methodology [supporting], Visualization [supporting], Writing—original draft [supporting]), Stephan von Gunten (Funding acquisition [equal], Methodology [supporting], Writing—review & editing [equal]), Jamie Heimburg-Molinaro (Conceptualization [equal], Methodology [equal], Project administration [lead], Writing—review & editing [lead]), and Richard Cummings (Conceptualization [lead], Funding acquisition [equal], Project administration [equal], Resources [lead], Supervision [lead], Writing—review & editing [equal]).

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Conflict of interest statement: None declared.

Data availability

The glycan maps for the Calendar heatmaps are provided as Supplementary Dataset S1 for the CFG array and Supplementary Dataset S2 for the NCFG array. Glycan sequences for the Lectin QA/QC array are provided as Supplementary Dataset S3. All GLAD files and data are available through the Harvard Dataverse: https://doi.org/10.7910/DVN/2GHR1D.

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